

et al., 2024b; De Bortoli et al., 2021) and continuous time (Shi et al., 2023; Peluchetti, 2023a; Vargas et al., 2021) settings, as well as their online versions (De Bortoli et al., 2024; Peluchetti, 2025; Karimi et al., 2024). Furthermore, the forward-backward IPMF framework could enable rectified flows to avoid error accumulation, making them even more powerful in generative modeling.

Notations. $\mathcal{P}_{2,ac}(\mathbb{R}^D)$ is a set of absolutely continuous distributions on $\mathbb{R}^D$ with finite second moment and finite entropy. We fix $N \geq 1$ intermediate time moments and set $0 = t_0 < t_1 < \dots < t_N < t_{N+1} = 1$. Let $q \in \mathcal{P}_{2,ac}(\mathbb{R}^{D \times (N+2)})$ be an associated discrete stochastic process on this grid. For any such q , we denote the density at $(x_0, x_{t_1}, \dots, x_{t_N}, x_1) \in \mathbb{R}^{D \times (N+2)}$ as $q(x_0, x_{\text{in}}, x_1)$, with $x_{\text{in}} = (x_{t_1}, \dots, x_{t_N})$. W^ϵ is a Wiener process with volatility $\epsilon > 0$ and initial distribution p_0 . Let p^{W^ϵ} be its discretization, i.e., $p^{W^\epsilon}(x_0, x_{\text{in}}, x_1) = p_0(x_0) \prod_{n=1}^{N+1} \mathcal{N}(x_{t_n} | x_{t_{n-1}}, \epsilon(t_n - t_{n-1}) I_D)$, where $\mathcal{N}(\cdot | \cdot)$ is a conditional Gaussian distribution. $H(q)$ is the differential entropy of q .

2 BACKGROUND

This section details the study’s key concepts; §2.1 introduces the Schrödinger Bridge (SB) problem, §2.2 presents the Iterative Proportional Fitting (IPF), §2.3 describes the Iterative Markovian Fitting (IMF), §2.4 discusses the heuristic modification of the IMF (Bidirectional IMF).

Recall that the SB problem (Schrödinger, 1931), IPF, and IMF admit both discrete- and continuous-time setups leading to the same problem solution. Moreover, the explicit formulas for IPF and IMF in the discrete setting are expressed in terms of probability densities, which helps to convey the main idea of our paper. Thus, for the sake of presentation flow, **the main text focuses exclusively on the discrete setup**, while Appendix C presents the continuous setup.

2.1 SCHRÖDINGER BRIDGE (SB) PROBLEM

The SB problem with a Wiener prior in the discrete-time setting (De Bortoli et al., 2021), given the initial distribution $p_0(x_0)$ and the final distribution $p_1(x_1)$, is stated as

$$\min_{q \in \Pi_N(p_0, p_1)} \text{KL}(q(x_0, x_{\text{in}}, x_1) || p^{W^\epsilon}(x_0, x_{\text{in}}, x_1)), \quad (1)$$

where $\Pi_N(p_0, p_1) \subset \mathcal{P}_{2,ac}(\mathbb{R}^{D \times (N+2)})$ is the subset of discrete stochastic processes with marginals $q(x_0) = p_0(x_0)$, $q(x_1) = p_1(x_1)$. The objective function in (1) admits a decomposition

$$\begin{aligned} \text{KL}(q(x_0, x_{\text{in}}, x_1) || p^{W^\epsilon}(x_0, x_{\text{in}}, x_1)) &= \text{KL}(q(x_0, x_1) || p^{W^\epsilon}(x_0, x_1)) \\ &+ \int \text{KL}(q(x_{\text{in}} | x_0, x_1) || p^{W^\epsilon}(x_{\text{in}} | x_0, x_1)) q(x_0, x_1) dx_0 dx_1. \end{aligned}$$

All $q(x_{\text{in}} | x_0, x_1)$ can be chosen independently of $q(x_0, x_1)$. Thus, we can consider $q(x_{\text{in}} | x_0, x_1) = p^{W^\epsilon}(x_{\text{in}} | x_0, x_1)$ and get $\text{KL}(q(x_{\text{in}} | x_0, x_1) || p^{W^\epsilon}(x_{\text{in}} | x_0, x_1)) = 0$.

This leads to the **Static SB problem**:

$$\min_{q \in \Pi(p_0, p_1)} \text{KL}(q(x_0, x_1) || p^{W^\epsilon}(x_0, x_1)), \quad (2)$$

where $\Pi(p_0, p_1) \subset \mathcal{P}_{2,ac}(\mathbb{R}^{D \times D})$ is the subset of joint distributions $q(x_0, x_1)$ s.t. $q(x_0) = p_0(x_0)$, $q(x_1) = p_1(x_1)$. Finally, the static SB objective can be expanded (Gushchin et al., 2023a, Eq. 7)

$$\text{KL}(q(x_0, x_1) || p^{W^\epsilon}(x_0, x_1)) = \int \frac{\|x_1 - x_0\|^2}{2\epsilon} dq(x_0, x_1) - H(q(x_0, x_1)) + C, \quad (3)$$

that is equivalent to the objective of the *entropic optimal transport* (EOT) problem with the *quadratic cost* up to an additive constant (Cuturi, 2013; Peyré et al., 2019; Léonard, 2013; Genevay, 2019).

2.2 ITERATIVE PROPORTIONAL FITTING (IPF)

Early works on SB (Vargas et al., 2021; De Bortoli et al., 2021; Tang et al., 2024) propose computational methods based on the IPF procedure (Kullback, 1968). The IPF-based algorithm is started by setting the process $q^0(x_0, x_{\text{in}}, x_1) = p_0(x_0)p^{W^\epsilon}(x_{\text{in}}, x_1 | x_0)$. Then, the algorithm alternates between two types of IPF projections, proj_1 and proj_0 , given by (De Bortoli et al., 2021, Prop. 2):

$$q^{2k+1} = \underbrace{\text{proj}_1 \left(q^{2k}(x_1) \prod_{n=0}^N q^{2k}(x_{t_n} | x_{t_{n+1}}) \right)}_{q^{2k}(x_1) q^{2k}(x_0, x_{\text{in}} | x_1)} \stackrel{\text{def}}{=} p_1(x_1) \underbrace{\prod_{n=0}^N q^{2k}(x_{t_n} | x_{t_{n+1}})}_{q^{2k}(x_0, x_{\text{in}} | x_1)}, \quad (4)$$